Structure & Syllabus for Semester-V

Integrated M.Sc. (Computer Science) Programme offered in

Department of Computer Science

Gujarat University

2025 – 2026

As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR UNDERGRADUATE PROGRAMMES, UGC

&

Resolution No. KCG/admin/2023-24/0607/kh.1

of

Education Department, Govt. of Gujarat

Semester - V

(Integrated M.Sc. (Computer Science) Programme)

MINORS

1. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (AIML)
2. INFORMATION SECURITY (IS)
3. WEB TECHNOLOGIES (WT)

Course Name: Data Analytics

Course Code: DSC-M-CS-354T

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1**: Introduce the necessary understanding of human intelligence and to explore the mechanisms that enables the intelligent thought and action
- **CO2:** Understand and learn effective ways for representing knowledge, applying intelligent problem-solving techniques & searching techniques

Prerequisites:

Basic knowledge of Python

Contents:

Unit No.	Particulars	Credits	Time
1.	**Unit Title: Introduction to Data Scales, Various Data Measures, Probability and Distributions:** *Introduction to Data Scales and Measures of Location:* Mean, Median, Mode, Percentiles, Quartiles; *Measures of Variability:* Range, Interquartile range, Variance, Standard deviation, Coefficient of variation; *Measures of Association between two variables:* Definition of Correlation, Types of Correlation, Methods of Correlation, Covariance interpretation of the covariance correlation, Coefficient interpretation of the correlation coefficient; *Measures of Distribution:* Shape, Relative location, and detecting outliers z-scores empirical rule detecting outliers box plot *Introduction to Probability:* Counting rules, Combinations and Permutations assigning probabilities, Events and their probabilities, Some basic relationships of probability, Complement of an event addition law, Conditional probability, Independent events multiplication law; *Probability Distributions:* Random Variables, Discrete vs Continuous random variable, *Discrete probability distribution:* Binomial, Poisson, Hyper geometric, *Continuous probability distribution:* Normal and standard normal distributions	**01**	**15 Hrs.**
2.	**Unit Title: Sampling, Sampling Distribution, Interval Estimation, Hypothesis Testing and Overview of Non-Parametric tests:** *Sampling, Sampling Distribution:* Random Sample Techniques, Non-random Sampling, Sampling and Non-sampling Errors, Sampling Distribution of Means, Sampling Distribution of Population, Properties of Point Estimators, *Interval Estimation:* Population Mean (σ: known & unknown), Determining Sample Size, Population Proportion *Hypothesis Test:* Null hypothesis, Alternative hypothesis, Simple and	**01**	**15 Hrs.**

Unit No.	Particulars	Credits	Time
	composite hypothesis, Type- I & type-II Errors, Test of significance of population Mean (σ: known & unknown), Level of significance, Determining test statistics, p-value approach, Critical value approach Determination of Sample size - (z-test, t-test), Overview of Non-Parametric tests		

References:

1. Anderson, Sweeney, Williams, "Statistics for Business and Economics", 12th edition, Cengage Publishers
2. Black, "Applied Business Statistics", 7th edition, Wiley
3. Aczel, Sounderpandian, Saravanan, Joshi, "Complete Business Statistics", 7th edition, McGraw-Hill
4. Levin, Rubin, "Statistics for Management", 8th edition, Pearson
5. David P. Doane, Lori E. Seward, "Applied Statistics in Business and Economics", 3rd edition, McGraw-Hill
6. Larry Wasserman, "All of Statistics", Springer
7. Richard A. Johnson, "Probability and Statistics for Engineers", 6th edition, PHI

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Develop strong analytical skills by working with real datasets
- Perform statistical tests (z-test, t-test), and interpreting results using software tools
- Derive meaningful insight through performing data visualization, probability applications, and hypothesis validation
- Apply statistical methods accurately in professional setups

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Course Name: Data Analytics – Practicals

Course Code: DSC-M-CS-354P

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1**: Understand key concepts of data, data mining, data analysis and data analytics
- **CO2:** Understand how data is created, stored, accessed
- **CO3:** Understand tools to effectively organize and visualize data
- **CO4:** Put up the principles and methods of statistical analysis into practice using a range of real-world data sets

Prerequisites:

Basic of Python

Contents:

Unit No.	Particulars	Credits	Time
1.	**Unit Title: Implementation of Various Data Measures, Probability Rules and Distributions** *Measures of Central Tendency, Measure of Dispersion, Measures of Association between two variables:* Perform computation programmatically, *Probability Concepts:* Demonstration of basic probability rules using random and collections, Basic relationships of probability, Conditional probability, *Probability Distributions:* Performing Discrete probability distributions and Continuous probability distributions programmatically	01	30 Hrs.
2.	**Unit Title: Computation of Interval Estimation and Hypothesis Testing** *Interval Estimation*: Population Mean (σ: known & unknown), *Hypothesis Test:* Test of significance of population Mean (σ: known & unknown)	01	30 Hrs.

References:

1. Anderson, Sweeney, Williams, "Statistics for Business and Economics", 12th edition, Cengage Publishers
2. Black, "Applied Business Statistics", 7th edition, Wiley
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